Operating Systems

Home Assignment - 03

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1) memory API and Types of memory.

A) memory API

memory API provides functions that allow program to allocate access and release memory during execution.

common memory operations

1) Allocate memory
2) Access memory
3) free memory

Types

1) cache memory.
2) main memory (RAM)
3) secondary memory.
4) virtual memory.

2) Logical Address to physical Address Translation.

A) A logical address is generated by the CPU, while a physical address refers to the actual location in memory.

Base Register -> contains the starting address limit Register of the process in physical memory.

-> contains the size of the process.

Example: Base register = 3000

logical address = 150

physical address = 3000+150 = 3150.

The limit register ensures the logical address does not exceed the allowed memory range.

Role

-> prevents a program from accessing memory of other programs.

3) free space management Techniques

A) free space management is used to track unused memory blocks.

1) splitting

when a large free block is allocated partially the remaining part becomes a new free block.

Ex: free block = 500kB

process = 200kB

allocated = 200kB

Remaining free = 300kB

2) Coalescing

adjacent free blocks are merges together to create a large block.

Ex: free block1 = 200kB

free block2 = 300kB

After coalescing = 500kB

4) fragmentation

fragmentation occurs when memory space is wasted or unusable.

Types of fragmentation

1) Internal fragmentation

occurs when allocated memory block is larger than the required memory.

Ex: Block size = 10kB

process size = 7kB

unused = 3kB.

2) External fragmentation:

occurs when free memory is divided into small scattered blocks.

Ex: free blocks = 5KB + 3KB + 2KB

process needs = 8KB

total free = 10KB but cannot allocate.

methods to reduce External fragmentation

1) compaction

2) paging

3) segmentation with paging.

5) memory Allocation Techniques

A) memory allocation is divided into contiguous and non-contiguous allocation.

contiguous memory Allocation.

-> process occupy continuous memory blocks.

Types

1) first fit

-> Allocate first suitable block.

2) Best fit

-) Allocates smallest block that fits

3) worst fit

-) Allocates largest block

Non-contiguous memory Allocation.

-) process is divided into multiple parts stored in different locations.

Techniques

1) paging

2) segmentation

3) paged segmentation.

6) Swapping in virtual memory

Definition

Swapping is a technique where a process is temporarily moved from main memory to secondary storage and later brought back for execution.

Goal of swapping

-) To increase CPU utilization.

-) To support multiprogramming.

advantages

1) allows more processes to run simultaneously.

2) improves memory utilization.

3) support large programs.

7) Thrashing and Degree of multiprogramming

Thrashing

Thrashing occurs when the system spends more time swapping pages than executing processes.

-> These happens when there are two many process in memory.

Degree of multiprogramming

number of process present in memory at a time.

* low multipro -> CPU Idle.

* optimal multiprogramming -> High CPU utilization.

* Top high multiprogramming -> Thrashing.

8) Translation look side Buffer (TLB)

-> TLB is a small high-speed cache memory that stores recent page table entries.

-> It helps in fast address translation.

working of TLB

1) CPU generates logical address

2) TLB checks if page number exists

3) If found -> TLB hit.

4) If not found -> TLB miss access page table.